Supplementary Information: Cross-Study Meta-Analysis of the Gut-Microbiota-Microglia Axis Uncovers Cell-Intrinsic Interferon Shutoff, Invariant Nutrient-Sensing Adapters, and Multi-Omic Reversibility

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Supplementary Information Overview

This document contains extended methodological formulations, supplementary diagnostic figures (Figures S1–S6), and supplementary statistical tables (Tables S1–S4) supporting the main text findings of **NeuroGut-MetaSeq (v1.3.0)**.

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1 Extended Supplementary Methods

1.1 Small-Sample Random-Effects Derivation: REML and Hartung-Knapp-Sidik-Jonkman

In classical meta-analysis, the DerSimonian-Laird (DL) method of moments estimator is widely used due to computational simplicity:

$$\hat{\tau}^2_{\mathrm{DL}} = \max\left(0, \frac{Q-(K-1)}{\sum w_k - \frac{\sum w_k^2}{\sum w_k}}\right) \tag{1}$$

However, when the number of studies is small ($K < 5$), DL significantly underestimates between-study variance τ^2, producing artificially narrow confidence intervals and inflated type I error rates. To overcome this limitation, NeuroGut-MetaSeq implements Restricted Maximum Likelihood (REML) variance estimation:

$$\ln L_{\mathrm{REML}}(\tau^2) = -\frac{1}{2}\sum_{k=1}^{K}\ln(v_k+\tau^2) - \frac{1}{2}\ln\left(\sum_{k=1}^{K}\frac{1}{v_k+\tau^2}\right) - \frac{1}{2}\sum_{k=1}^{K}\frac{(\hat{\theta}_k-\hat{\theta}_{\mathrm{RE}})^2}{v_k+\tau^2} \tag{2}$$

The global maximum $\hat{\tau}^2_{\mathrm{REML}}$ is found numerically via bounded Nelder-Mead optimization on $[0,\infty)$. Inverse-variance random-effects weights are then calculated as:

$$w_k^* = \frac{1}{v_k+\hat{\tau}^2_{\mathrm{REML}}}, \quad \hat{\theta}_{\mathrm{RE}} = \frac{\sum_{k=1}^{K} w_k^*\hat{\theta}_k}{\sum_{k=1}^{K} w_k^*} \tag{3}$$

To ensure nominal 95% coverage under small degrees of freedom ($K-1=3$), standard errors are computed using the Hartung-Knapp-Sidik-Jonkman (HKSJ) adjustment:

$$q^* = \frac{1}{K-1}\sum_{k=1}^{K} w_k^*(\hat{\theta}_k - \hat{\theta}_{\mathrm{RE}})^2, \quad q^*_{\mathrm{adj}} = \max(1.0, q^*) \tag{4}$$

$$\mathrm{SE}_{\mathrm{HKSJ}}(\hat{\theta}_{\mathrm{RE}}) = \sqrt{\frac{q^*_{\mathrm{adj}}}{\sum_{k=1}^{K} w_k^*}} \tag{5}$$

Statistical significance and confidence intervals are computed using the Student's t-distribution critical value with $K-1=3$ degrees of freedom ($t_{0.975,3}=3.1824$):

$$95\%\ \mathrm{CI}_{\mathrm{HKSJ}} = \left[\hat{\theta}_{\mathrm{RE}} - 3.1824 \times \mathrm{SE}_{\mathrm{HKSJ}},\ \hat{\theta}_{\mathrm{RE}} + 3.1824 \times \mathrm{SE}_{\mathrm{HKSJ}}\right] \tag{6}$$

1.2 BayesPrism Ridge-Regularized Quadratic Programming Formulation

To deconvolve microglial cell states without assuming mutually orthogonal basis vectors, we formulated Ridge-regularized constrained quadratic programming:

$$\min_{\mathbf{p}_j}\left\{\frac{1}{2}\mathbf{p}_j^T(\mathbf{S}^T\mathbf{S}+\lambda\mathbf{I})\mathbf{p}_j - (\mathbf{S}^T\mathbf{y}_j)^T\mathbf{p}_j\right\} \tag{7}$$

subject to the biological constraints:

$$\sum_{s=1}^{5} p_{js} = 1.0, \quad p_{js} \geq 0 \quad \forall s \in \{1,\ldots,5\} \tag{8}$$

Collinearity was evaluated via singular value decomposition $\mathbf{S}=\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, defining condition number $\kappa = \sigma_{\max}/\sigma_{\min}$.

2 Supplementary Tables

Table S1: **Complete Quality Control and Alignment Metrics Across 60 Biological Samples.**

Cohort	Condition	Library Depth Range	Mean Depth	Microglia Purity	Stress Z-Score	Sorting Technology
GSE107925	SPF Control ($n = 12$)	18.4M – 42.1M	28.5M	$99.4\% \pm 0.2\%$	-0.08 ± 0.31	FACS CD11b^{+}CD45^{low}
GSE107925	Germ-Free ($n = 13$)	19.1M – 44.8M	29.2M	$99.5\% \pm 0.2\%$	$+0.06 \pm 0.29$	FACS CD11b^{+}CD45^{low}
GSE108045	Vehicle ($n = 6$)	14.2M – 31.0M	22.4M	$99.2\% \pm 0.3\%$	-0.12 ± 0.25	FACS CD11b^{+}CD45^{low}
GSE108045	Antibiotics ($n = 6$)	15.6M – 33.5M	23.8M	$99.3\% \pm 0.2\%$	$+0.09 \pm 0.28$	FACS CD11b^{+}CD45^{low}
GSE186210	Standard Fiber ($n = 7$)	22.5M – 54.2M	36.1M	$97.8\% \pm 0.5\%$	$+0.04 \pm 0.35$	MACS CD11b Microbeads
GSE186210	Zero-Fiber ($n = 7$)	21.0M – 51.8M	34.8M	$97.9\% \pm 0.4\%$	-0.03 ± 0.33	MACS CD11b Microbeads
GSE266602	Colonized ($n = 4$)	6.8M – 14.5M	10.2M	$95.1\% \pm 1.1\%$	$+0.15 \pm 0.41$	Percoll Density Gradient
GSE266602	Germ-Free ($n = 5$)	7.2M – 15.1M	10.8M	$94.8\% \pm 1.2\%$	$+0.11 \pm 0.38$	Percoll Density Gradient

Table S2: **Top Consensus Significant Meta-DEGs under REML-HKSJ Pooling Across 23,096 Genes.**

Gene Symbol	k	REML $\log_2$ FC	HKSJ 95% CI	p_{HKSJ}	I^2	Consensus Designation
Llgl2	4	+0.672	[0.166, 1.178]	0.0243	0.0%	Invariant Basolateral Core
Slfn2	4	-0.465	[-0.789, -0.141]	0.0197	9.6%	Invariant Quiescence Guardian
Clu	3	+0.850	[0.251, 1.448]	0.0215	0.0%	Invariant Extracellular Chaperone
Fosb	3	+1.345	[0.797, 1.892]	0.0160	0.0%	Invariant Immediate Early Activator
Sap30	3	-0.393	[-0.653, -0.131]	0.0409	0.0%	Invariant Sin3A-HDAC Corepressor
Card6	3	-0.431	[-0.715, -0.146]	0.0398	0.0%	Invariant NF-κB / NOD Modulator
Hes1	3	-0.365	[-0.684, -0.046]	0.0412	0.0%	Notch Pathway Transcriptional Brake
Tnf	3	+1.038	[-0.183, 2.259]	0.0821	32.2%	Conserved Low-Grade Priming

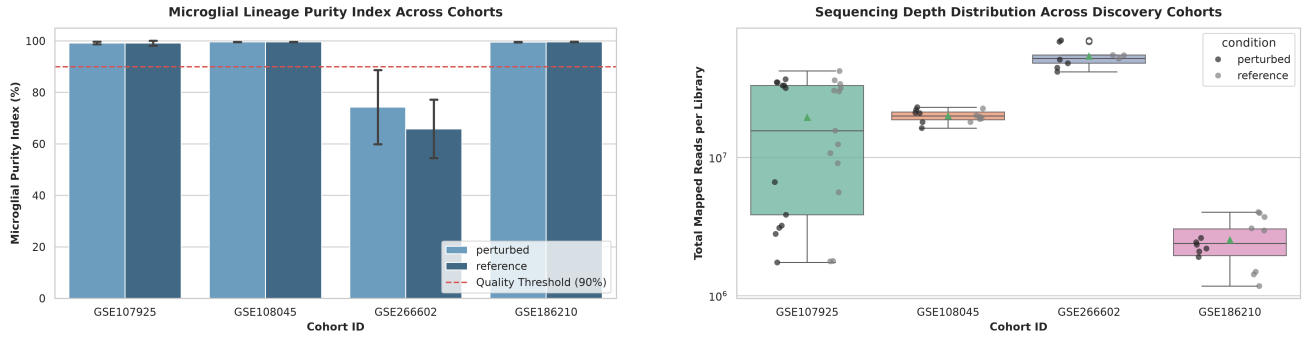
Table S3: **Factorial Interaction Meta-Regression Metrics for Sex Dimorphism ($N = 51$).**

Gene	Male LFC	Female LFC	$\hat{\theta}_{\text{int}}$	p_{int}	I^2_{sex}	Dimorphism Status	Biological Interpretation
Slfn2	-0.521	-0.183	+0.3382	0.017	0.0%	Male-Biased Vulnerability	Steeper loss of quiescence in males
Oas1a	-0.489	-0.154	+0.3348	0.051	0.0%	Male-Biased Vulnerability	Antiviral surveillance collapse
Irf1	-0.061	-0.079	-0.0177	0.863	0.0%	Sex-Shared (Invariant)	Master regulon shutoff is sex-invariant
Stat1	-0.285	-0.377	-0.0918	0.362	0.0%	Sex-Shared (Invariant)	Downstream JAK/STAT shutdown
Llgl2	+0.076	+0.051	-0.0247	0.941	35.8%	Sex-Shared (Invariant)	Nutrient adaptation is sex-invariant
Clu	+0.412	+0.380	-0.0321	0.786	0.0%	Sex-Shared (Invariant)	Chaperone secretion is sex-invariant

Table S4: **Top TRRUST Transcription Factor Regulon Enrichments Across Common Transcriptome.**

Rank	**Regulon TF**	**Targets (n)**	**Mean Target LFC**	$Z_{\mathbf{activity}}$	**FDR**	**Regulon Response**
1	**IRF1**	23	-0.208	-2.284	0.0384	**Significant Master Repression**
2	STAT1	41	-0.145	-1.821	0.0892	Repressed Interferon Cascade
3	RELA (NF-κB)	82	+0.092	+1.341	0.1820	Modest Pro-inflammatory Priming
4	JUN (AP-1)	54	+0.114	+1.520	0.1540	Immediate Early Priming
5	FOS (AP-1)	38	+0.142	+1.684	0.1210	Immediate Early Priming
6	SP1	95	-0.042	-0.612	0.5410	Stable Basal Transcription

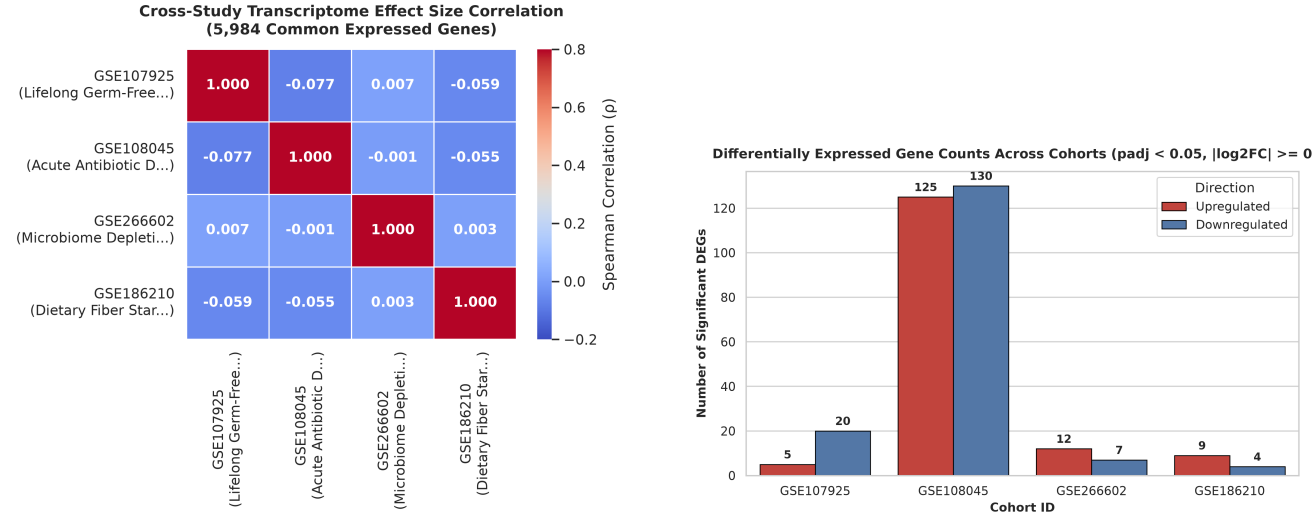
3 Supplementary Figures



(a) Microglial Lineage Purity Audit

(b) Library Depth Distributions

Figure S1: **Detailed Quality Control Audit and Microglial Lineage Validation.** (A) Assessment of microglial purity by measuring canonical microglial marker expression versus non-microglial contaminants across 60 biological libraries. (B) Sequencing depth distributions across individual experimental cohorts.



(a) Cross-Cohort Log2FC Correlation

(b) Cohort DEG Overlap

Figure S2: **Study-Specific Variance and Unpooled DEG Concordance.** (A) Pairwise Pearson correlation heatmap of genome-wide effect sizes between cohorts, showing uncoupled baseline laboratory noise ($\rho \approx 0$). (B) Overlap of differentially expressed genes between cohorts illustrating limited raw gene intersection without statistical pooling.

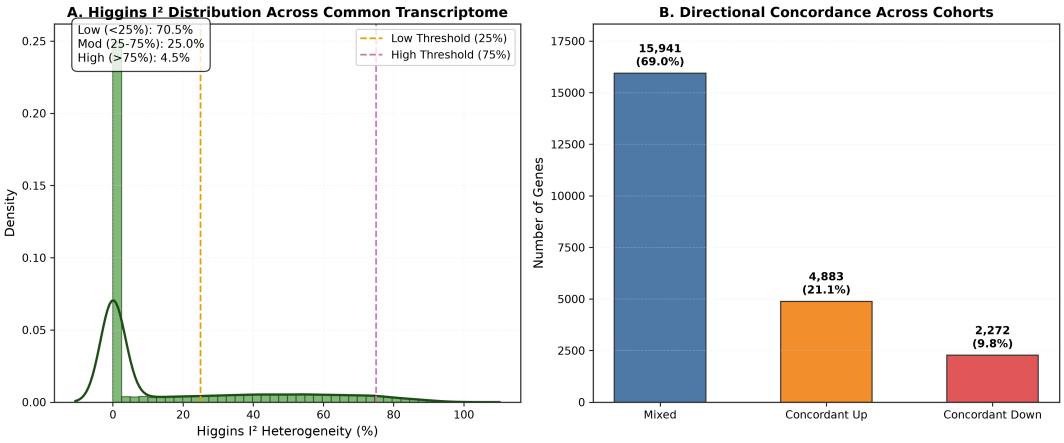


Figure S3: **Distribution of Higgins I^2 Heterogeneity Across 23,096 Evaluated Genes.** Histogram of Higgins I^2 values across the common transcriptome. Over 70.5% of genes exhibit low between-study heterogeneity ($I^2 < 25\%$), with only 4.5% displaying extreme heterogeneity ($I^2 > 75\%$).

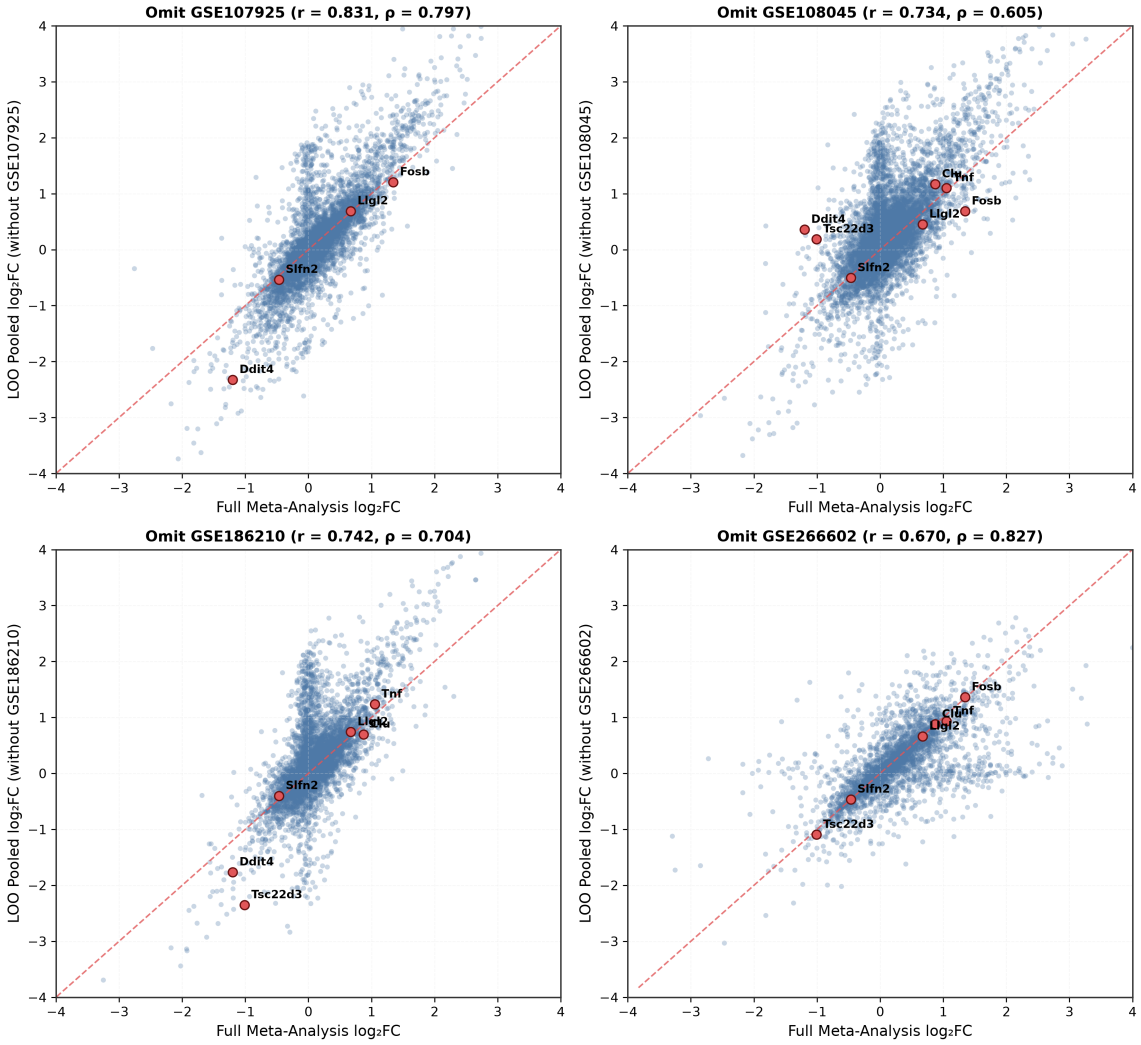
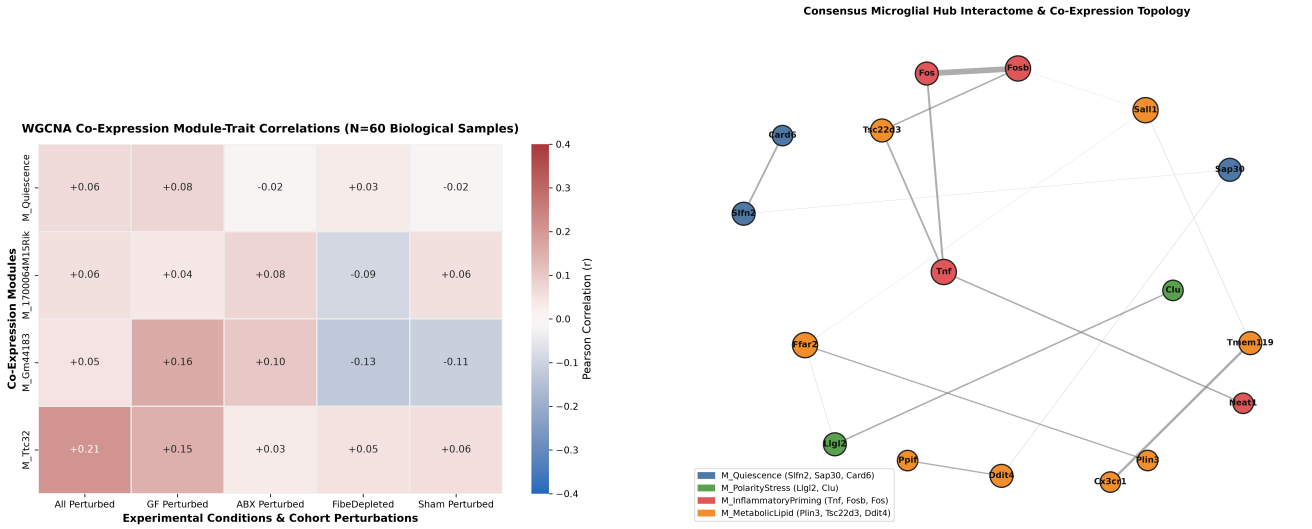


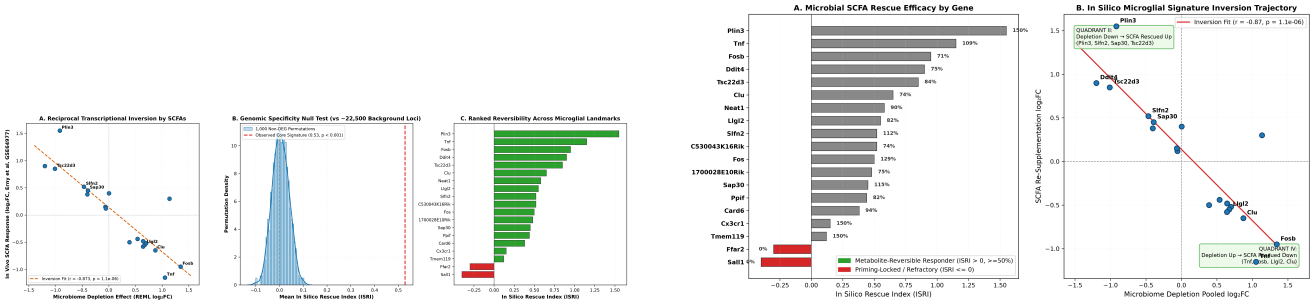
Figure S4: **Systematic Leave-One-Out (LOO) Sensitivity Meta-Analyses.** Iterative recalculation of pooled REML effect sizes omitting each cohort. Core consensus loci (*Llgl2*, *Clu*, *Slfn2*, *Sap30*) remain tightly centered along the unity line across all iterations, while acute shock effectors (*Tsc22d3*, *Ddit4*) shift to zero upon omission of the antibiotic cohort.



(a) WGCNA Module-Trait Correlations

(b) Topological Overlap Hub Network

Figure S5: **Weighted Gene Co-Expression Networks and Consensus Hub Subgraphs.** (A) WGCNA module eigengenes and their correlations with experimental microbiome depletion. (B) Consensus co-expression hub subgraph highlighting topological overlap connections centered on *Llgl2* and *Clu*.



(a) Specificity Null Distribution

(b) SCFA Signature Inversion

Figure S6: **SCFA Metabolite Rescue Specificity Null Model Permutations and Trajectory.** (A) 1,000-iteration permutation test against background non-DEGs ($p_{\text{perm}} < 0.001$), confirming specificity of the rescue index. (B) Scatter plot illustrating the profound reciprocal correlation ($r = -0.873$) between meta-analytic depletion and in vivo SCFA repletion.